Software Requirement Specifications

Orbis AI - Intelligent Travel Planning System

Version 1.5

Final Year Project

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Supervisor	Ms. Bakhtawar Abbasi
Co-Supervisor	—
Project Team	M. Talha Yousif (Leader) Salik Ahmed Bhatti Toheed Ali
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National University of Computer & Emerging Sciences
FAST-NUCES

Document History

This section maintains a comprehensive revision history to track all changes made to the document.

Version	Author	Date	Description of Change
1.0	M. Talha Yousif	29/11/2025	Initial document creation and structure setup
1.1	M. Talha Yousif	29/11/2025	Introduction section completed with objectives
1.2	Salik Ahmed Bhatti	29/11/2025	Project scope and exclusions defined
1.3	Toheed Ali	29/11/2025	System constraints and interface requirements
1.4	M. Talha Yousif	30/11/2025	Use case diagram and functional/non-functional requirements
1.5	Project Team	30/11/2025	References added and comprehensive document review

Distribution List

The following individuals will receive this document after each official sign-off.

Name	Role
Ms. Bakhtawar Abbasi	Project Supervisor
—	Co-Supervisor
M. Talha Yousif	Project Leader
Salik Ahmed Bhatti	Team Member
Toheed Ali	Team Member

Document Sign-Off

This section records the official approval of the document by the designated authorities.

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1.5	Supervisor	30/11/2025

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1 Introduction

1.1 Purpose of Document

This Software Requirements Specification (SRS) document provides a comprehensive and structured outline of the functional and non-functional requirements for the **Orbis AI** system. It serves as the definitive reference for all project stakeholders throughout the system's development lifecycle.

Key Objectives:

- Establish a clear communication framework among all stakeholders
- Serve as the primary reference for system design, implementation, and testing
- Ensure alignment between system implementation and business objectives
- Provide a foundation for requirement traceability throughout the project

1.2 Intended Audience

This document is designed for the following stakeholders:

- **Academic Stakeholders:** FAST-NUCES administration, jury members, and students
- **Project Team:** Designers, developers, and testers
- **Supervision:** Ms. Bakhtawar Abbasi (Project Supervisor)
- **End Users:** Potential users and early adopters of Orbis AI

1.3 Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
ICS	iCalendar Standard
LLM	Large Language Model
MARL	Multi-Agent Reinforcement Learning
NLP	Natural Language Processing
OAuth	Open Authorization
RAG	Retrieval-Augmented Generation
UX	User Experience

1.4 Document Conventions

- **Font Family:** Arial
- **Font Size:** 12pt for headings, 10-11pt for body text
- **Color Coding:** Blue for primary headings, gray for secondary information
- **Emphasis:** Bold text for key terms and important concepts

2 Overall System Description

2.1 Project Background

Modern travel planning suffers from significant fragmentation. Users must navigate multiple platforms for flights, accommodations, and activities, manually reconcile pricing and policies, and risk missing optimal options due to information overload.

Problem Statement

- **Fragmented Services:** Inefficient planning across multiple disconnected platforms
- **Lack of Personalization:** Limited real-time adaptation to user preferences
- **Complexity:** Difficult comparison and booking across providers
- **Trust Issues:** Need for transparent, ethical AI in handling sensitive user data

Orbis AI Solution: An autonomous travel agent leveraging agentic AI to understand natural language queries, decompose them into structured tasks, and integrate results from real-time APIs to deliver personalized, optimized travel itineraries.

2.2 Project Scope

Orbis AI aims to revolutionize travel planning through intelligent AI orchestration, providing transparency, efficiency, and deep personalization.

2.2.1 Included Functionalities

- **Conversational Interface:** Natural language-based travel planning via chat
- **AI Orchestration:** Intelligent coordination of flights, hotels, and itineraries
- **API Integration:** Amadeus, Booking.com, Stripe sandbox, Google Maps
- **Personalization Engine:** Preference learning through questionnaires and embeddings
- **Real-Time Support:** Dynamic trip management, notifications, and change handling
- **Sandbox Payments:** Demo-only transactions with no real charges
- **MARL Optimization:** Multi-Agent Reinforcement Learning for optimal planning
- **Multi-Modal Summaries:** LLM-powered trip visualization with text, maps, and images
- **Dynamic Re-Planning:** Real-time constraint solver for itinerary adjustments
- **Future Features:** Admin panel for analytics, logs, and system audits

2.2.2 Excluded Functionalities

- Car rentals and travel insurance services
- Corporate travel management and compliance rules
- Production-grade ticketing with full regulatory compliance

2.3 Not In Scope

- Real-time API response verification beyond sandbox environments
- Production-scale user load handling
- Manual booking verification or full production compliance

2.4 Project Objectives

2.4.1 User Functions

1. Initiate trip planning through natural language conversations
2. Receive AI-curated travel bundles (flights, hotels, activities)
3. Interactively refine itineraries based on personal preferences
4. Execute test bookings with sandbox payment processing

5. Export itineraries to calendar format (ICS)
6. Receive automated email notifications
7. Access real-time support for trip modifications

2.4.2 Success Criteria

Core Success Metrics:

- User successfully signs in and initiates trip request
- System generates curated flight and hotel bundles
- Realistic day-by-day itinerary created within budget constraints
- Test payment processed with booking confirmation
- ICS calendar export successfully generated
- Change requests processed with updated options

Advanced Features:

- **MARL Implementation:** Reinforcement learning agents dynamically negotiate between flight, hotel, and itinerary agents to optimize cost, travel time, and user satisfaction
- **Multi-Modal Summaries:** LLM-powered generation combining text, annotated maps, and landmark images for rich trip visualization
- **Real-Time Optimization:** Constraint solver maintains feasible itineraries during dynamic changes (delays, weather, preference updates)

2.4.3 Admin Functions (Future)

- Access observability dashboards for agent execution monitoring
- Audit user interactions and system performance metrics

2.5 Stakeholders

Stakeholder	Interest
Travelers	End users seeking simplified travel planning
Development Team	System designers, developers, and testers
Supervisor	Academic oversight and guidance
FAST-NUCES	Institutional stakeholder
API Providers	Amadeus, Booking.com, Stripe, Google
Potential Investors	Future commercialization opportunities

2.6 Operating Environment

- **Platform:** Web-based application with internet connectivity requirement
- **Browsers:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
- **APIs:** Sandbox mode API keys for all third-party integrations
- **Deployment:** Vercel/Railway with database connectivity
- **Database:** SurrealDB or PostgreSQL with pgvector extension

2.7 System Constraints

Technical Constraints

- **Database:** SurrealDB or PostgreSQL
- **AI Frameworks:** LangChain, Microsoft AutoGen
- **Third-Party APIs:** Amadeus, Booking.com, Stripe (sandbox), Google Maps
- **Network Dependency:** All features require stable internet connectivity
- **API Limitations:** Subject to rate limits and potential downtime

Design Constraints

- **Language:** English as primary interface language
- **Ethical AI:** Bias mitigation in recommendation algorithms
- **UX Requirements:** Intuitive interface for diverse user backgrounds

2.8 Assumptions & Dependencies

2.8.1 Assumptions

- Users provide accurate preferences and travel constraints
- Sandbox API behavior mirrors production environments
- Users possess basic web navigation skills
- System used for demonstration purposes only (no real payments)

2.8.2 Dependencies

- **External APIs:** Availability and rate limit compliance
- **Frontend Framework:** Next.js with React and TypeScript
- **Caching:** Redis for performance optimization
- **Authentication:** NextAuth with OAuth 2.0 support

3 External Interface Requirements

3.1 Hardware Interfaces

Component	Requirement
Desktop/Laptop	Minimum 4GB RAM, modern processor
Mobile Devices	iOS 13+, Android 8.0+
Internet Connection	Minimum 2 Mbps recommended
Display	Minimum 1280×720 resolution

3.2 Software Interfaces

Component	Description
Database System	SurrealDB/PostgreSQL for persistent storage
Web Framework	Next.js for frontend and backend
Flight API	Amadeus Travel APIs for flight search
Hotel API	Booking.com Distribution API
Payment Gateway	Stripe (sandbox mode)
Mapping Service	Google Maps API
Authentication	NextAuth (Google OAuth + Email/Password)
Email Service	Resend/SendGrid for notifications
Vector Store	pgvector/Elasticsearch for embeddings

3.3 Communications Interfaces

- **Data Format:** JSON for API data exchange
- **Real-Time:** WebSockets for chat functionality
- **Security:** HTTPS/TLS 1.3 for all communications
- **Synchronization:** Automatic real-time data sync

4 Functional Requirements

4.1 Functional Hierarchy

System Architecture Overview

The Orbis AI system is organized into seven major functional modules, each containing specialized sub-functions for comprehensive travel planning capabilities.

4.1.1 Module 1: User Management

- 1.1 User Registration
- 1.2 User Login (OAuth/Email)
- 1.3 Profile Management
- 1.4 Preference Viewing

4.1.2 Module 2: Trip Planning

- 2.1 Chat Session Initiation
  - 2.1.1 Cold-Start Questionnaire
  - 2.1.2 Natural Language Query Processing
- 2.2 Itinerary Generation
  - 2.2.1 Flight Search
  - 2.2.2 Hotel Search
  - 2.2.3 Activity Recommendations
- 2.3 Itinerary Refinement
- 2.4 Real-Time Support
- 2.5 Multi-Modal Summary Generation

4.1.3 Module 3: Booking Management

- 3.1 Bundle Confirmation
- 3.2 Test Payment Processing (Sandbox)
- 3.3 Option Holding
- 3.4 Change Request Handling

4.1.4 Module 4: Personalization

- 4.1 Preference Learning (RAG)
- 4.2 Embeddings Update

4.1.5 Module 5: Notifications & Exports

- 5.1 Email Notifications
- 5.2 ICS Calendar Export
- 5.3 Trip Reminders

4.1.6 Module 6: Agentic AI

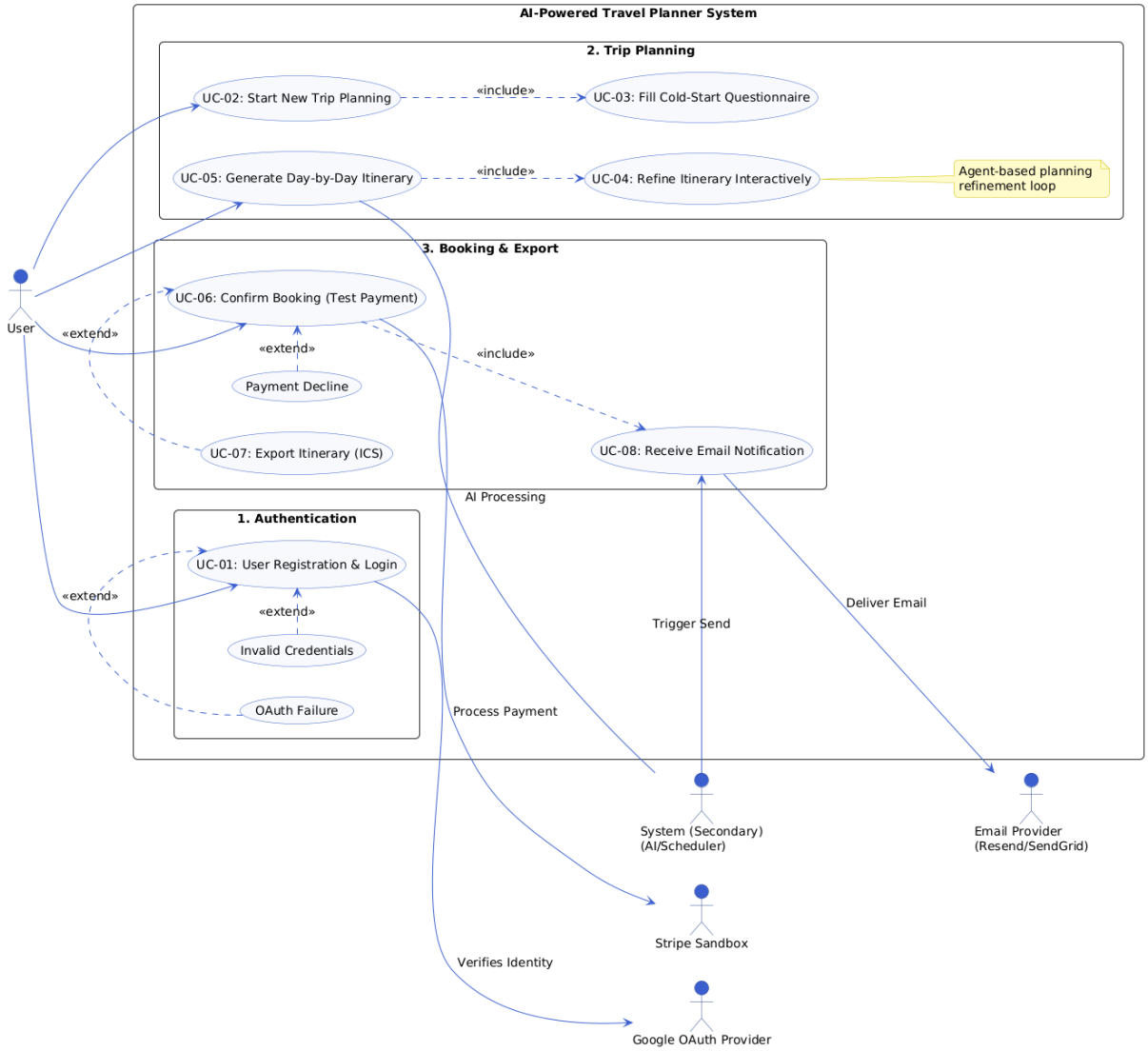
- 6.1 Planner Agent
- 6.2 Flight Agent
- 6.3 Hotel Agent
- 6.4 Itinerary Agent
- 6.5 Booking Agent
- 6.6 Verifier Agent
- 6.7 MARL Optimization
- 6.8 Constraint Solver

4.1.7 Module 7: Admin Panel (Future)

- 7.1 Log Viewing
- 7.2 Audit Dashboard

4.2 Use Cases

4.2.1 Use Case Diagram



4.2.2 UC-01: User Registration & Login

Use Case Specification	
Use Case ID	UC-01
Actors	User (Primary)
Feature	User Registration and Authentication via Google OAuth or Email/Password
Preconditions	• User has valid Google account or email address • System is online and accessible • Authentication service is operational

Main Scenario:

Step	User Action	System Response
1	User navigates to login page	Displays authentication options (OAuth, Email)
2	User selects authentication method	Redirects to OAuth provider or displays login form
3	User submits credentials	Validates credentials and authenticates user
4	N/A	Creates user profile if new user

Alternate Scenarios:

- **3a - Invalid Credentials:** System displays error message with retry option
- **4a - Existing User:** System logs user in directly to dashboard
- **2a - OAuth Failure:** System handles provider errors gracefully and falls back to alternative methods
- **Network Issue:** System displays connection error and suggests retry

Postconditions:

- User is authenticated and session is established
- User is redirected to personalized dashboard
- User preferences are loaded if existing

Cross-Referenced Use Cases:

- UC-02: Start New Trip Planning
- UC-03: Fill Cold-Start Questionnaire
- UC-11: Update User Preferences

4.2.3 UC-02: Start New Trip Planning (Chat)

Use Case Specification	
Use Case ID	UC-02
Actors	User (Primary)
Feature	Initiate trip planning via conversational chat interface
Preconditions	• User is authenticated • No active trip planning session exists • APIs are available in sandbox mode

Main Scenario:

Step	User Action	System Response
1	User opens chat interface	Displays welcome message and prompt for trip details
2	User enters natural language query (e.g., ”Plan a trip to Paris for 5 days”)	Planner agent parses query and initiates specialized agents
3	N/A	Generates initial travel bundles using integrated APIs
4	N/A	Presents options with personalized recommendations

Alternate Scenarios:

- **2a - Ambiguous Query:** System asks clarifying questions
- **3a - API Failure:** System uses cached data or notifies user of delay

Postconditions:

- Trip planning session is active
- Initial itinerary bundles are displayed
- User preferences are applied

Cross-Referenced Use Cases:

- UC-03: Fill Cold-Start Questionnaire
- UC-04: Refine Itinerary Interactively
- UC-12: Multi-City Flight Search

4.2.4 UC-03: Fill Cold-Start Questionnaire

Use Case Specification	
Use Case ID	UC-03
Actors	User (Primary)
Feature	Complete initial preference questionnaire for personalization
Preconditions	• New user or no preferences set • User is authenticated

Main Scenario:

Step	User Action	System Response
1	N/A	System prompts questionnaire upon first login
2	User responds to questions (budget, interests, etc.)	Validates responses in real-time
3	User submits questionnaire	Stores responses in vector store
4	N/A	Generates initial embeddings for personalization

Alternate Scenarios:

- **2a - Skip Question:** System applies default values
- **3a - Incomplete Submission:** System saves partial data and allows later completion

Postconditions:

- User preferences are saved
- RAG system is initialized for user

Cross-Referenced Use Cases:

- UC-01: User Registration & Login
- UC-11: Update User Preferences

4.2.5 UC-04: Refine Itinerary Interactively

Use Case Specification	
Use Case ID	UC-04
Actors	User (Primary)
Feature	Interactively modify and refine proposed itinerary
Preconditions	• Initial itinerary generated • Active planning session

Main Scenario:

Step	User Action	System Response
1	User requests modification (e.g., ”Change hotel to cheaper option”)	Itinerary agent analyzes request
2	N/A	Requeries APIs for updated options
3	N/A	Presents revised bundles with explanations
4	User confirms changes	Updates session state

Alternate Scenarios:

- **1a - Invalid Request:** System suggests valid alternatives
- **2a - Constraint Violation:** Notifies user of conflicts (e.g., budget exceed)

Postconditions:

- Itinerary is updated
- Changes are persisted in session

Cross-Referenced Use Cases:

- UC-02: Start New Trip Planning
- UC-05: Generate Day-by-Day Itinerary
- UC-09: Request Trip Change

4.2.6 UC-05: Generate Day-by-Day Itinerary

Use Case Specification	
Use Case ID	UC-05
Actors	User (Primary)
Feature	Generate detailed day-by-day travel itinerary
Preconditions	• Travel bundles selected • Preferences applied

Main Scenario:

Step	User Action	System Response
1	User requests itinerary generation	Itinerary agent compiles data
2	N/A	Integrates activities and routes via Google Maps
3	N/A	Applies MARL for optimization
4	N/A	Displays detailed day-by-day plan

Alternate Scenarios:

- **3a - Optimization Failure:** Falls back to rule-based planning
- **2a - No Activities:** Suggests additions based on preferences

Postconditions:

- Complete itinerary generated
- Ready for booking or further refinement

Cross-Referenced Use Cases:

- UC-04: Refine Itinerary Interactively
- UC-06: Confirm Booking with Test Payment
- UC-13: Hotel Comparison

4.2.7 UC-06: Confirm Booking with Test Payment

Use Case Specification	
Use Case ID	UC-06
Actors	User (Primary)
Feature	Confirm travel bundle with sandbox payment processing
Preconditions	• Itinerary finalized • Stripe sandbox configured

Main Scenario:

Step	User Action	System Response
1	User selects confirm booking	Redirects to secure checkout page
2	User enters test payment details	Processes transaction in sandbox mode (no real charge)
3	N/A	Generates booking confirmation
4	N/A	Triggers email notification and ICS export

Alternate Scenarios:

- **2a - Payment Decline:** Simulates decline and allows retry
- **1a - Hold Option:** Temporarily reserves without payment

Postconditions:

- Booking confirmed in demo mode
- All documents generated

Cross-Referenced Use Cases:

- UC-05: Generate Day-by-Day Itinerary
- UC-07: Export Itinerary to Calendar (ICS)
- UC-08: Receive Email Notification

4.2.8 UC-07: Export Itinerary to Calendar (ICS)

Use Case Specification	
Use Case ID	UC-07
Actors	User (Primary)
Feature	Export confirmed itinerary to ICS calendar format
Preconditions	• Booking confirmed • Itinerary details available

Main Scenario:

Step	User Action	System Response
1	User selects export option	Generates ICS file from itinerary data
2	User downloads or adds to calendar	Provides download link or direct integration
3	N/A	Confirms successful export

Alternate Scenarios:

- **1a - Format Error:** Validates and regenerates file
- **2a - Calendar Integration Fail:** Provides manual import instructions

Postconditions:

- ICS file generated and available
- Itinerary events added to user's calendar

Cross-Referenced Use Cases:

- UC-06: Confirm Booking with Test Payment
- UC-05: Generate Day-by-Day Itinerary

4.2.9 UC-08: Receive Email Notification

Use Case Specification	
Use Case ID	UC-08
Actors	User (Primary), System (Secondary)
Feature	Send automated email notifications for trip updates
Preconditions	• Event trigger (e.g., booking confirmation) • User email verified

Main Scenario:

Step	User Action	System Response
1	N/A	Detects notification trigger event
2	N/A	Composes personalized email with details
3	N/A	Sends via email provider (Resend/SendGrid)
4	User receives email	Logs delivery status

Alternate Scenarios:

- **3a - Delivery Failure:** Retries send or logs error
- **2a - Custom Template:** Uses different templates for event types

Postconditions:

- Notification delivered
- User informed of updates

Cross-Referenced Use Cases:

- UC-06: Confirm Booking with Test Payment
- UC-09: Request Trip Change

4.2.10 UC-09: Request Trip Change

Use Case Specification	
Use Case ID	UC-09
Actors	User (Primary)
Feature	Process real-time changes to existing trip plans
Preconditions	• Active or confirmed trip exists • Change constraints feasible

Main Scenario:

Step	User Action	System Response
1	User submits change request (e.g., ”Shift departure by 2 days”)	Constraint solver evaluates impact
2	N/A	Re-plans itinerary using MARL agents
3	N/A	Presents updated options with cost differences
4	User approves	Applies changes and notifies via email

Alternate Scenarios:

- **2a - Infeasible Change:** Suggests closest alternatives
- **1a - External Trigger:** Handles automatic changes (e.g., flight delay)

Postconditions:

- Trip updated successfully
- New itinerary persisted

Cross-Referenced Use Cases:

- UC-04: Refine Itinerary Interactively
- UC-08: Receive Email Notification

4.2.11 UC-10: View Past Trips

Use Case Specification	
Use Case ID	UC-10
Actors	User (Primary)
Feature	Access and review history of planned trips
Preconditions	• User has completed trips • Authenticated session

Main Scenario:

Step	User Action	System Response
1	User navigates to history section	Retrieves trips from database
2	User selects specific trip	Displays detailed itinerary and bookings
3	User requests export	Generates summary or ICS file

Alternate Scenarios:

- **1a - No History:** Displays message ”No past trips available”
- **2a - Data Error:** Handles gracefully with retry option

Postconditions:

- Trip details displayed
- Optional export completed

Cross-Referenced Use Cases:

- UC-01: User Registration & Login
- UC-07: Export Itinerary to Calendar (ICS)

4.2.12 UC-11: Update User Preferences

Use Case Specification	
Use Case ID	UC-11
Actors	User (Primary)
Feature	Modify and update personal travel preferences
Preconditions	• User authenticated • Profile exists

Main Scenario:

Step	User Action	System Response
1	User accesses profile settings	Displays current preferences
2	User modifies preferences	Validates changes in real-time
3	User saves updates	Updates vector store and embeddings
4	N/A	Applies changes to active sessions

Alternate Scenarios:

- **2a - Invalid Input:** Displays validation errors
- **3a - Partial Save:** Saves valid changes only

Postconditions:

- Preferences updated
- Future recommendations adjusted

Cross-Referenced Use Cases:

- UC-03: Fill Cold-Start Questionnaire
- UC-01: User Registration & Login

4.2.13 UC-12: Multi-City Flight Search

Use Case Specification	
Use Case ID	UC-12
Actors	User (Primary)
Feature	Perform flight search for multi-city itineraries
Preconditions	• Active planning session • Multi-city route specified

Main Scenario:

Step	User Action	System Response
1	User specifies multi-city details	Flight agent queries Amadeus API
2	N/A	Filters results based on preferences
3	N/A	Displays comparison table
4	User selects option	Integrates into itinerary

Alternate Scenarios:

- **1a - No Results:** Suggests alternative routes
- **2a - Price Filter:** Applies budget constraints

Postconditions:

- Flight segments added
- Itinerary updated

Cross-Referenced Use Cases:

- UC-02: Start New Trip Planning
- UC-05: Generate Day-by-Day Itinerary

4.2.14 UC-13: Hotel Comparison

Use Case Specification	
Use Case ID	UC-13
Actors	User (Primary)
Feature	Compare hotel options based on multiple criteria
Preconditions	• Destination specified • Dates available

Main Scenario:

Step	User Action	System Response
1	User requests hotel options	Hotel agent queries Booking.com API
2	N/A	Applies filters (price, location, ratings)
3	N/A	Displays comparison matrix
4	User selects hotel	Adds to itinerary bundle

Alternate Scenarios:

- **1a - No Availability:** Suggests alternatives or date changes
- **3a - Custom Sort:** Allows user-defined sorting

Postconditions:

- Hotel selected
- Bundle updated

Cross-Referenced Use Cases:

- UC-05: Generate Day-by-Day Itinerary
- UC-04: Refine Itinerary Interactively

4.2.15 UC-14: Generate Multi-Modal Summary

Use Case Specification	
Use Case ID	UC-14
Actors	User (Primary)
Feature	Generate visual and textual trip summary using LLM
Preconditions	• Itinerary complete • Multi-modal APIs available

Main Scenario:

Step	User Action	System Response
1	User requests summary	LLM processes itinerary data
2	N/A	Integrates maps, images, and annotations
3	N/A	Displays interactive multi-modal view
4	User interacts	Updates view dynamically

Alternate Scenarios:

- **2a - API Unavailable:** Falls back to text-only summary
- **3a - Export Request:** Generates shareable format

Postconditions:

- Summary generated
- Enhanced user understanding

Cross-Referenced Use Cases:

- UC-05: Generate Day-by-Day Itinerary
- UC-07: Export Itinerary to Calendar (ICS)

4.2.16 UC-15: Admin View Observability Dashboard (Future)

Use Case Specification	
Use Case ID	UC-15
Actors	Admin (Primary)
Feature	View system logs and performance traces
Preconditions	• Admin authenticated • Dashboard enabled

Main Scenario:

Step	User Action	System Response
1	Admin accesses dashboard	Displays real-time metrics and logs
2	Admin applies filters	Queries and shows filtered data
3	Admin views agent traces	Displays detailed run information
4	Admin exports report	Generates downloadable summary

Alternate Scenarios:

- **1a - Unauthorized Access:** Denies entry and logs attempt
- **2a - No Data:** Displays ”No recent activity” message

Postconditions:

- Dashboard viewed successfully
- System insights gained

Cross-Referenced Use Cases:

- UC-01: User Registration & Login

5 Non-functional Requirements

5.1 Performance Requirements

Performance Metrics

- **Concurrency:** Support up to 10 simultaneous users without degradation
- **Response Time:** Itinerary generation within 5 seconds
- **API Latency:** External calls complete in under 10 seconds average
- **AI Processing:** Orchestration results in 20 seconds or less
- **Transaction Time:** Sandbox confirmations within 15 seconds
- **Uptime:** 95% availability for demonstration environments

5.2 Safety Requirements

- Implement automated database backups to prevent data loss
- Robust error handling to maintain system stability
- Schema validation to prevent AI hallucinations
- Compliance with ethical data handling practices (GDPR-inspired)
- Fallback mechanisms for API failures

5.3 Security Requirements

Security Controls

- **Authentication:** Google OAuth and email/password with secure hashing
- **Encryption:** Standard encryption for sensitive data (API keys, preferences)
- **Access Control:** Role-based for future admin features
- **Communications:** Mandatory HTTPS for all interactions
- **Privacy:** No data sharing without explicit user consent

5.4 User Documentation

- Comprehensive User Manual covering all features
- Integrated Online Help with contextual tooltips
- Video Tutorials for key workflows (e.g., chat interface)
- Context-Sensitive Guidance for complex operations
- Troubleshooting Guide for common issues

6 References

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7 Appendices

This section includes supplementary materials that support the main document content.

- **Appendix A:** Detailed System Architecture Diagram
- **Appendix B:** API Integration Specifications
- **Appendix C:** Data Models and Schemas
- **Appendix D:** Test Case Templates